OS A2 Student Evaluation Sheet

Name: ____________________ Section: ____________ Roll No: _______________

Q1) Multiplayer Terminal Game - Zombie Survival (Total: 50 Marks)

Criteria	Description	Max Marks	Self-Evaluation	Marks Obtained
Game Mechanics & Concept	Basic game structure: player, zombies, manager, actions.	10		
Process Management	Correct use of fork(), exec() and process control.	10		
Inter-Process Communication	Functional use of pipes between all entities.	6		
Unique Student Requirement	Zombie count based on roll number logic.	5		
Docker Containerization	Working Dockerfile and ability to run the game containerized.	10		
Documentation	Clear README on how to run, structure, and features.	4		
Output & Gameplay	Proper terminal output showcasing working game flow.	5		

Q2) Professor-Student Synchronization (Total: 50 Marks)

Criteria	Description	Max Marks	Self-Evaluation	Marks Obtained
Basic Synchronization (Single Student)	One student arrives, asks a question, and waits for the professor's reply.	5		
Multiple Students Asking Sequentially	Ensures each student waits for the professor's answer before next starts.	15		
Professor Napping Feature	Professor sleeps when no student is present and wakes up properly.	10		
Multiple Students Competing (Concurrency)	Only one student asks at a time when multiple arrive together.	10		
Fairness and Order Maintenance	Students are answered in arrival order, no skips or repeats.	10		
Total		50		

Q3) Cricket Match Analysis Tool with Scheduling Algorithms (Total: 40 Marks)

Criteria	**Description**	**Max Marks**	**Self-Evaluation**	**Marks Obtained**
SJF Implementation	Prioritize players based on strike/economy rates for optimal order.	8		
SRTF Logic	Dynamically reorder players based on batting or bowling performance.	8		
Round Robin Logic	Use RR for bowler rotation, ensuring efficient workload management.	6		
Performance Simulation	Simulate match scenarios with created player data (batting/bowling).	8		
Analysis Report	Provide clear output and recommendations based on the simulated match.	6		

Code Quality & Structure	Code is clean, modular, and logically structured for readability and maintainability.	4		

Q4) Four-Process Circular Communication System (Total: 40 Marks)

Criteria	Description	Max Marks	Self-Evaluation	Marks Obtained
Process Creation (fork)	Four child processes are correctly created.	4		
Pipe Setup (IPC)	Unnamed pipes are properly connected in a circular pattern.	4		
Circular Communication Logic	Message flows A → B → C → D → A correctly.	6		
Message Modification	Each process appends its tag to the message.	6		
File Logging	Uses system calls to log modified message into log files.	6		
Message Display in Process A	Final output displayed by Process A after full cycle.	4		
Error Handling	Proper checks for pipe, fork, read/write, and open/close.	4		
Synchronization / Clean Exit	Clean termination and correct message ordering.	6		
Total		40		

Q5) Multi-Level Feedback Queue Scheduling (Total: 20 Marks)

Criteria	Description	Max Marks	Self-Evaluation	Marks Obtained
Correct Queue Implementation (MFQ Logic)	Implements 3-level feedback queue with correct algorithms and time quanta.	6		
Process Handling & Input Processing	Handles input correctly, including edge cases.	4		
Scheduling & CPU Time Management	Queue priority respected, correct transitions and preemptions.	4		
Turnaround Time & Waiting Time	Correct calculations and output formatting.	4		
Code Optimization & Readability	Efficient structures, modular, and clean code.	2		
Total		20		